EXPERIMENT 15
HASH FUNCTION MDS

AIM

To implement the **MD5 hash function** in C and generate the message digest for a given input message.

ALGORITHM

1. Start the program.
2. Read the input message.
3. Initialize the MD5 context.
4. Pass the message to the MD5 algorithm.
5. Finalize the hash calculation.
6. Convert the digest into hexadecimal form.
7. Display the MD5 hash value.
8. Stop the program.

CODE

```
#include <stdio.h>

#include <stdint.h>

#include <string.h>

#include <stdlib.h>

#define LEFTROTATE(x, c) (((x) << (c)) | ((x) >> (32 - (c))))

uint32_t K[64] = {

    0xd76aa478, 0xe8c7b756, 0x242070db, 0xc1bdceee,

    0xf57c0faf, 0x4787c62a, 0xa8304613, 0xfd469501,

    0x698098d8, 0x8b44f7af, 0xffff5bb1, 0x895cd7be,

    0x6b901122, 0xfd987193, 0xa679438e, 0x49b40821,

    0xf61e2562, 0xc040b340, 0x265e5a51, 0xe9b6c7aa,
```

```
    0xd62f105d, 0x02441453, 0xd8a1e681, 0xe7d3fbc8,
    0x21e1cde6, 0xc33707d6, 0xf4d50d87, 0x455a14ed,
    0xa9e3e905, 0xfcefa3f8, 0x676f02d9, 0x8d2a4c8a,
    0xfffa3942, 0x8771f681, 0x6d9d6122, 0xfde5380c,
    0xa4beea44, 0x4bdecfa9, 0xf6bb4b60, 0xbebfbc70,
    0x289b7ec6, 0xeaa127fa, 0xd4ef3085, 0x04881d05,
    0xd9d4d039, 0xe6db99e5, 0x1fa27cf8, 0xc4ac5665,
    0xf4292244, 0x432aff97, 0xab9423a7, 0xfc93a039,
    0x655b59c3, 0x8f0ccc92, 0xffeff47d, 0x85845dd1,
    0x6fa87e4f, 0xfe2ce6e0, 0xa3014314, 0x4e0811a1,
    0xf7537e82, 0xbd3af235, 0x2ad7d2bb, 0xeb86d391
};
uint32_t S[64] = {
    7,12,17,22, 7,12,17,22, 7,12,17,22, 7,12,17,22,
    5,9,14,20, 5,9,14,20, 5,9,14,20, 5,9,14,20,
    4,11,16,23, 4,11,16,23, 4,11,16,23, 4,11,16,23,
    6,10,15,21, 6,10,15,21, 6,10,15,21, 6,10,15,21
};
void md5(const unsigned char *msg, size_t len, unsigned char digest[16])
{
    uint32_t a0 = 0x67452301;
    uint32_t b0 = 0xefcdab89;
```

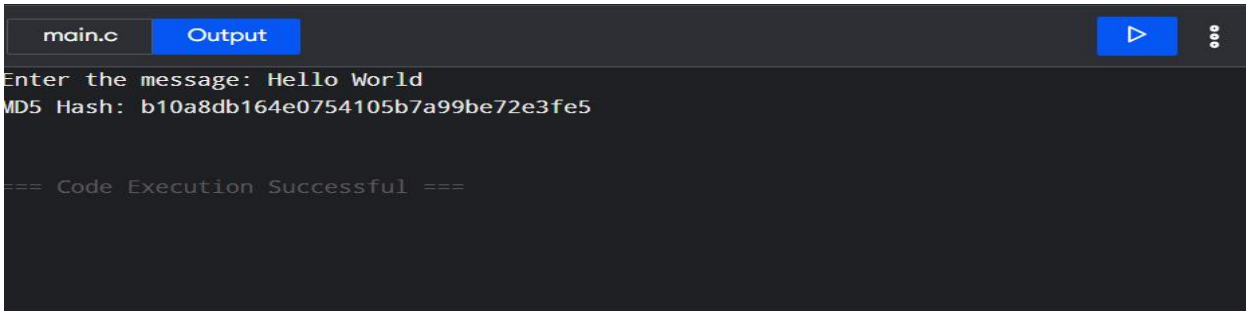
```
uint32_t c0 = 0x98badcfe;
uint32_t d0 = 0x10325476;
size_t new_len = len + 1;
while (new_len % 64 != 56)
    new_len++;
unsigned char *buffer = calloc(new_len + 8, 1);
memcpy(buffer, msg, len);
buffer[len] = 0x80;
uint64_t bits = (uint64_t)len * 8;
memcpy(buffer + new_len, &bits, 8);
for (size_t offset = 0; offset < new_len; offset += 64)
{
    uint32_t M[16];
    for (int i = 0; i < 16; i++)
    {
        M[i] =
            (uint32_t)buffer[offset + i * 4] |
            ((uint32_t)buffer[offset + i * 4 + 1] << 8) |
            ((uint32_t)buffer[offset + i * 4 + 2] << 16) |
            ((uint32_t)buffer[offset + i * 4 + 3] << 24);
    }
    uint32_t A = a0;
```

```
uint32_t B = b0;
uint32_t C = c0;
uint32_t D = d0;
for (int i = 0; i < 64; i++)
{
    uint32_t F;
    int g;
    if (i < 16)
    {
        F = (B & C) | ((~B) & D);
        g = i;
    }
    else if (i < 32)
    {
        F = (D & B) | ((~D) & C);
        g = (5 * i + 1) % 16;
    }
    else if (i < 48)
    {
        F = B ^ C ^ D;
        g = (3 * i + 5) % 16;
    }
```

```
        else
        {
            F = C ^ (B | (~D));
            g = (7 * i) % 16;
        }
        uint32_t temp = D;
        D = C;
        C = B;
        B = B + LEFTROTATE(
            A + F + K[i] + M[g],
            S[i]
        );
        A = temp;
    }
    a0 += A;
    b0 += B;
    c0 += C;
    d0 += D;
}
memcpy(digest, &a0, 4);
memcpy(digest + 4, &b0, 4);
memcpy(digest + 8, &c0, 4);
```

```
    memcpy(digest + 12, &d0, 4);
    free(buffer);
}
int main()
{
    char message[1000];
    unsigned char digest[16];
    printf("Enter the message: ");
    fgets(message, sizeof(message), stdin);
    message[strcspn(message, "\n")] = '\0';
    md5((unsigned char *)message, strlen(message), digest);
    printf("MD5 Hash: ");
    for (int i = 0; i < 16; i++)
        printf("%02x", digest[i]);
    printf("\n");
    return 0;
}
```

OUTPUT



RESULT

Thus, the **MD5 hash function** was successfully implemented in C and the message digest was generated.